AIFFELTHON · SPATIAL AVLA · POC BASELINE

AVLA: Audio–Vision–Language Alignment for Event Triggering

A LangGraph DAG that turns siren detection, DOA tracking and visual anomaly into one VLM-ready context frame

Contribution — a contract-first, schema-validated pipeline (S_I → [S_A ‖ S_T ‖ S_V] → S_B → S_C) with a 2×2 live operator UI, measured end-to-end on real dashcam and real siren recordings.

3.06	0	0.827	0.70
ms p50 UI fast path (target < 33 ms)	schema errors · All_OK = True	siren head accuracy (mixed clips)	trigger rate on real sirens

Problem — A siren is an event you must hear *and* see › **AVLA PoC** — LangGraph DAG · JSON-Schema contract · hysteresis › **Operator** — 2×2 multi-view · MONITOR / CAUTION / ALERT

00 · In action The 2×2 operator view (real run)

Four live tiles — camera stream, siren head timeline, vision risk and the VLM payload — rendered by the Gradio app from a stitched dashcam clip. The bottom-right tile is the operator recommendation (**MONITOR / CAUTION / ALERT_OPERATOR**); the app issues no control command.

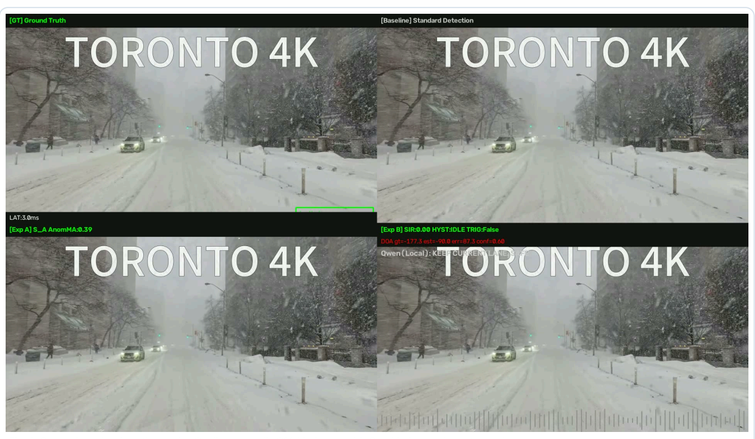


Fig. 0. Screenshot taken from the running Gradio app on clip C01 (day) — a real 2×2 multi-view render, not a mock-up.

01 · Problem One event, three modalities

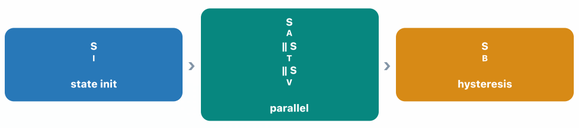
An emergency siren in traffic is **audible** (direction of arrival), **visible** (surrounding vehicles, weather) and must become **language** an operator can act on.

Single-modality detectors fail exactly where it matters: a siren can be masked by rain noise, and a visual anomaly can be a false positive on a tunnel wall.



Fig. 1. S_A stage: RT-DETR-L on a real dashcam frame (C01, day) — 41 detections. Real checkpoint, real frames.

02 · System LangGraph DAG, six agents



S_A visual anomaly (RT-DETR-L + MAE) · S_T siren head (WavLM) · S_V verification fixture · S_B multimodal fusion with a **35-frame hysteresis** · S_C VLM payload builder.

Contract-first

Every stage exchanges one **FrameAudioPacket** validated against a Draft-7 JSON Schema — **pre** and **final**. A schema violation is an error, not a warning.

03 · Measured Latency and contract health

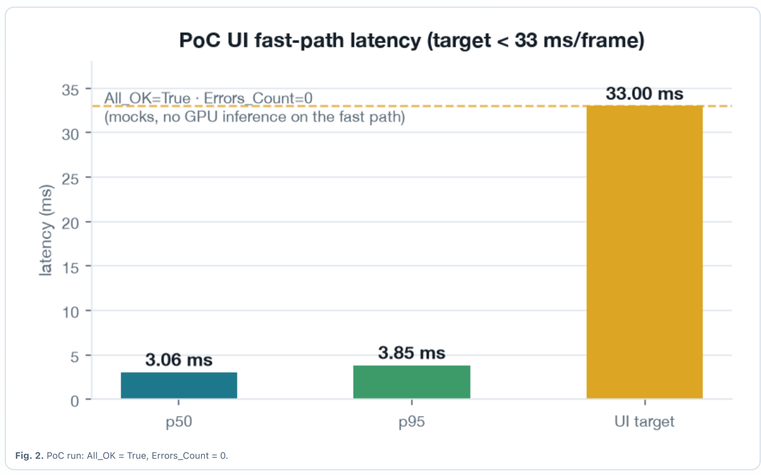


Fig. 2. PoC run: All_OK = True, Errors_Count = 0.

Audio–video sync verified on 4 clips (T09, C01, C02, C04): 450 frames each, 16 kHz / 30 fps clock, **exact sample match**.

04 · Vision (S_A) RT-DETR-L + MAE anomaly

Risk score = RT-DETR-L detection evidence combined with a MAE embedding distance to a normal clip. The **mock baseline is a constant 0.20**; the detector must clear 0.75 to trigger.

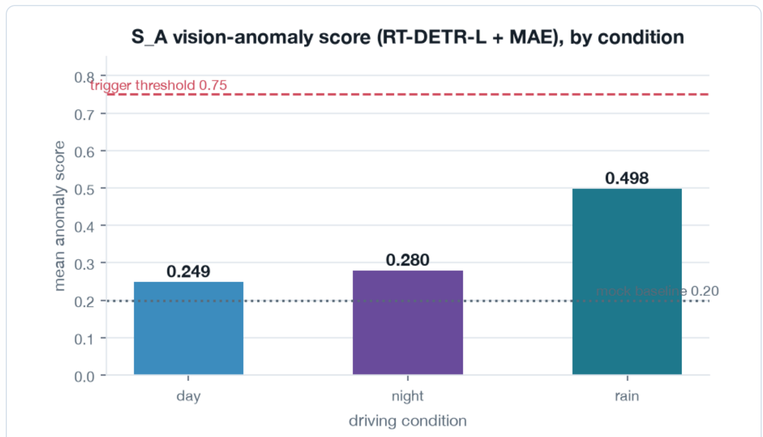


Fig. 3. Mean anomaly score by condition (RT-DETR-L, MPS).

Rain (C02) roughly **doubles** the day score — the detector reacts to a wet tunnel wall. Cost: **242–301 ms/frame**, so this stage runs off the UI fast path.

05 · Audio (S_B) WavLM head + hysteresis

WavLM-base-plus embeddings → L2-logistic head. Head accuracy **0.827** on the mixed clip set; the pipeline triggers on **0.70** of real-siren frames and **0.027** of ordinary day traffic.

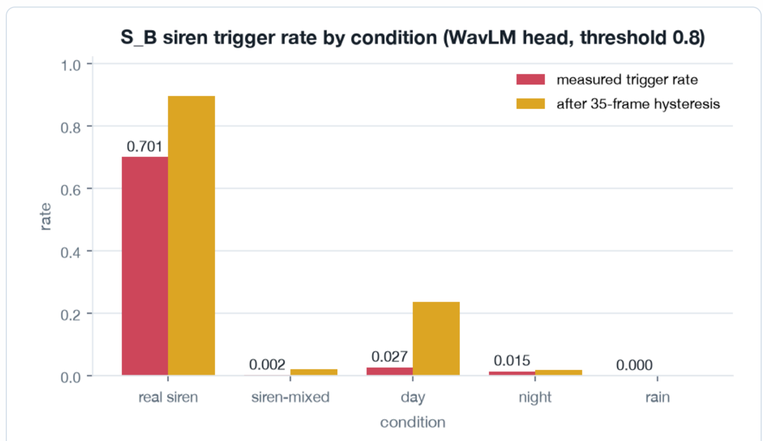


Fig. 4. Measured trigger rate, with and without the 35-frame hysteresis.

06 · Why a real siren head Synthetic baselines mislead

The first head was trained against a **synthesised** siren centroid. Measuring against **real** G44 sirens shows the synthetic baseline does not represent them: cosine to the synthetic centroid is **lower for real sirens** than for siren-mixed dashcam audio.



Fig. 5. WavLM frame cosine to the synthetic centroid.

Fix: train the head on **real** sirens — final head reaches **100%** test accuracy with >99.8% confidence on C01, after restoring the best epoch (13 of 30, early stop patience 3).

07 · Head quality Where the errors are

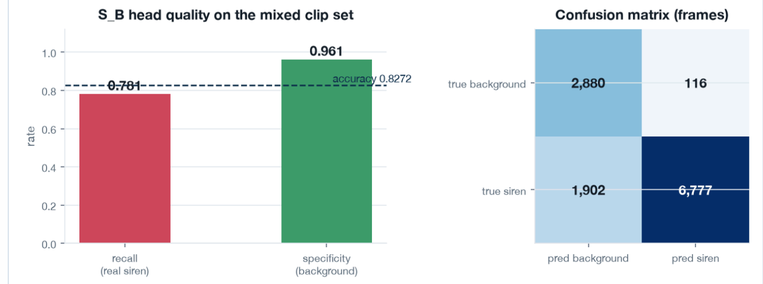


Fig. 6. S_B head: recall 0.781, specificity 0.961, accuracy 0.827 on the mixed set.

Misses dominate (fn 1902 vs fp 116) because the negative clips are long; the **hysteresis** converts isolated positives into a held event.

08 · Operating point Choosing the threshold

The AST template path exposes a tunable trade-off; the default **balanced** mode keeps every true positive with no false trigger on the 5 negative clips.

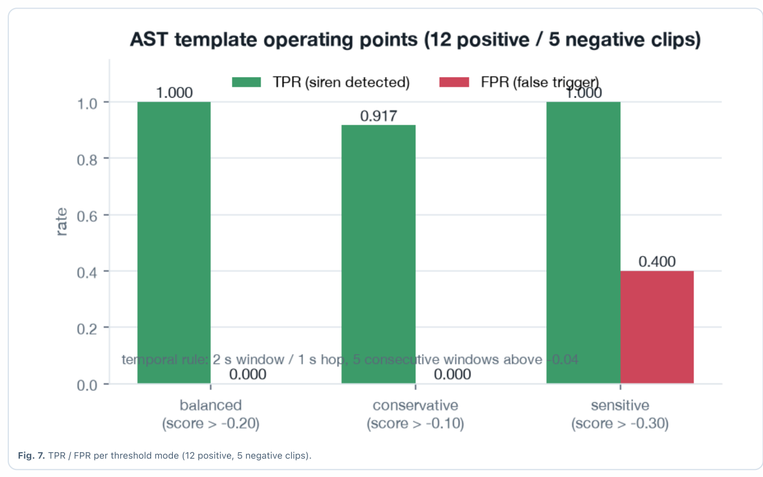


Fig. 7. TPR / FPR per threshold mode (12 positive, 5 negative clips).

Temporal rule: 2 s window / 1 s hop, and **5 consecutive** windows above −0.04. The untouched tunnel clip (T09) crossed the single-frame threshold 6/14 times but never 5 in a row — the streak rule removes exactly that false positive.

08b · VLM payload (S_C) From evidence to language

S_C turns the fused event into a **Qwen2.5-0.5B-Instruct** payload run as a separate background task, surfaced as **MONITOR / CAUTION / ALERT_OPERATOR** only.

Inference is rate-limited to **once per 5 s** and re-armed only when the class or input changes; a Qwen failure is shown in the status panel rather than swallowed. The payload is a recommendation for a human — never a command.

09 · Honest limits What this PoC does not prove

- The AST template clips (**G44-01/02/03, C01/C02**) are part of the evaluation set — **not an independent test**.
- Trigger rates are **provisional operating points**, not generalisation performance.
- S_A vision runs **242–301 ms/frame** (RT-DETR-L) and is **off** the 33 ms fast path.
- S_B clip-level scores read the **whole clip** audio — not causal real-time inference.
- PoC stages are **mock logic**; the app **drives no vehicle** and produces **no control output**.

09b · Evaluation What was measured, on what

Real dashcam: C01 (day), C02 (rain / Seoul tunnel), C04 (night), T09 (untouched tunnel) — 450–451 frames each at 30 fps, 16 kHz stereo audio.

Real sirens: 6 G44 recordings (air-raid, ambulance, warbling boat, emergency) — 8 600+ frames, used to train and score the head.

Audio template set: 12 positive / 5 negative clips, 2 s windows with 1 s hop; **sync check:** 4 clips verified frame-exact.

10 · Next From PoC to deployment

Replace synthetic siren baselines with a **real, held-out** siren set, move RT-DETR-L onto the live path, and port the pipeline to **3D detection** (DETR3D → PETR → Sparse4D) for the visual stage.

11 · Stack What it is built from

Python 3.10 · **LangGraph** DAG · **RT-DETR-L** (Ultralytics, MPS) · **WavLM-base-plus** (audio embeddings) · **MAE ViT-Tiny** (anomaly embedding) · **Qwen2.5-0.5B-Instruct** (VLM payload) · **Gradio** 2×2 multi-view UI · Draft-7 JSON-Schema contracts · pytest suite.

12 · Safety posture What the UI refuses to do

Recommendations are limited to **MONITOR / CAUTION / ALERT_OPERATOR**. The app issues **no throttle, steering or braking command**, and the live monitor reports the MPC mock's steering/brake state as **simulation-only** values that never reach an actuator.

12b · Reproducibility Everything is re-runnable

pytest suite over the contracts and agents · **golden fixtures** for every stage · Colab bundles for GPU inference (Qwen Audio + AST) · JSON-Schema contracts versioned as `contracts/frame_audio_packet_v1.0.json` · mock state shared by CLI and Gradio.

13 · Takeaway Contract first, hysteresis second, evidence third

Schema-validated packets make every stage auditable; hysteresis turns noisy frame scores into held events; and measuring against **real** sirens — not a synthetic stand-in — is what makes the audio head trustworthy.

Repo: spatial-avla-pipeline · **Report:** docs/AVLA_PoC_Final_Report.pdf